

Capital Funding Prudence and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Capital Funding Prudence (CFP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CFP achieves an annualized gross (net) Sharpe ratio of 0.57 (0.49), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (19) bps/month with a t-statistic of 2.56 (2.32), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets) is 18 bps/month with a t-statistic of 2.17.

1 Introduction

Asset pricing models fundamentally seek to explain cross-sectional variation in expected returns through differences in systematic risk exposure. Production-based asset pricing theory provides a compelling framework for understanding these differences by linking firms' real investment decisions to their cost of capital and expected returns. Despite substantial theoretical advances connecting production economies to asset prices, empirical evidence remains mixed on whether production-based factors fully explain the cross-section of stock returns. We identify Capital Funding Prudence (CFP) as a novel characteristic that captures firms' conservative approach to external financing relative to their investment needs. This measure reflects fundamental frictions in capital markets that affect firms' marginal costs of production and their exposure to aggregate productivity shocks, yet has received limited attention in the asset pricing literature.

In a production-based economy, firms' expected returns depend critically on their exposure to shocks affecting the marginal product of capital and investment adjustment costs [Cochrane and Piazzesi \(2005\)](#). Capital Funding Prudence captures firms' reliance on internal versus external financing, which directly maps to the shadow cost of external finance and investment frictions in production models. When firms exhibit high CFP, they face lower external financing frictions and can more readily adjust their capital stock in response to productivity shocks, leading to lower conditional risk premia [Zhang \(2005\)](#). The production-based framework predicts that firms with different levels of CFP will have heterogeneous exposures to aggregate productivity shocks. High-CFP firms, maintaining conservative funding policies, possess greater financial flexibility to exploit positive productivity shocks without incurring substantial adjustment costs [Cooper and Haltiwanger \(2006\)](#). These firms effectively smooth their marginal costs of production across states of the world, reducing their systematic risk. Conversely, low-CFP firms face binding financing constraints that

amplify their exposure to aggregate shocks, as external financing becomes particularly costly during economic downturns when the marginal utility of consumption is high [Livdan et al. \(2009\)](#). The cross-sectional variation in CFP therefore proxies for differences in firms’ operating leverage and their ability to hedge against adverse productivity shocks. In equilibrium, this heterogeneity in production technologies and financing frictions generates predictable variation in expected returns, with low-CFP firms commanding higher risk premia to compensate investors for bearing greater systematic risk related to aggregate productivity and investment shocks [Berk et al. \(1999\)](#).

We find strong evidence that Capital Funding Prudence predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on CFP achieves an annualized gross Sharpe ratio of 0.57 and a net Sharpe ratio of 0.49 after accounting for transaction costs. The strategy delivers a monthly average abnormal gross return of 22 basis points relative to the Fama-French five-factor model plus momentum (FF6), with a t-statistic of 2.56. The economic magnitude remains substantial after controlling for transaction costs, with net abnormal returns of 19 basis points per month and a t-statistic of 2.32. The predictive power of CFP proves robust across alternative portfolio construction methodologies. When we employ equal-weighted portfolios, name-based breakpoints, or market capitalization breakpoints, the strategy continues to generate significant abnormal returns with t-statistics exceeding 2.0 in most specifications. Importantly, the CFP effect persists among large-cap stocks, where the strategy earns 20 basis points per month with a t-statistic of 1.90 for firms above the 80th NYSE size percentile, suggesting the anomaly is not confined to illiquid small stocks. The strategy’s gross monthly alpha remains economically significant at 18 basis points with a t-statistic of 2.17 when we simultaneously control for the six Fama-French factors and six closely related anomalies including net external financing, asset growth, and changes in net operating assets. This evi-

dence indicates that CFP captures unique information about expected returns not subsumed by existing factors or related firm characteristics.

Our study makes several important contributions to the production-based asset pricing literature. First, we extend the empirical evidence on investment-based anomalies by introducing Capital Funding Prudence as a novel predictor that directly captures firms' external financing frictions, complementing existing work on investment and asset growth [Cooper et al. \(2008\)](#); [Fama and French \(2015\)](#). Unlike broad measures of investment or asset growth that conflate multiple economic mechanisms, CFP specifically isolates the role of funding constraints in production decisions, providing cleaner identification of the risk channel. Second, we demonstrate that CFP's predictive power survives controls for closely related anomalies documented in [Titman et al. \(2004\)](#) and [Lyandres et al. \(2008\)](#), suggesting it captures distinct variation in firms' production technologies and risk exposures not reflected in existing characteristics. Our evidence supports theoretical models linking external finance frictions to cross-sectional return variation through their effect on firms' operating leverage and systematic risk exposure. Third, our analysis reveals that production-based factors alone cannot fully explain the CFP premium, as significant alphas persist relative to the Fama-French five-factor model that includes profitability and investment factors [Fama and French \(2015\)](#). This finding suggests that standard implementations of investment-based factors may not adequately capture the heterogeneity in financing frictions across firms. The broader implications extend beyond anomaly documentation to inform theoretical development in production-based asset pricing. The persistence of the CFP premium highlights the importance of modeling heterogeneous financing frictions and their interaction with productivity shocks in generating cross-sectional return dispersion, consistent with recent theoretical advances in [Ai et al. \(2013\)](#).

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on firms with sufficient data to construct year-over-year changes in financing activities and non-missing property, plant, and equipment values. Capital Funding Prudence signal relies on two key variables from firms' financial statements. Financing activities net cash flow (FINCF) represents the net cash provided by or used in financing activities as reported in the statement of cash flows. This item captures cash flows from transactions with the firm's owners and creditors, including proceeds from issuing debt or equity, repayments of borrowed amounts, dividend payments, and treasury stock transactions. Property, plant, and equipment gross (PPEGT) measures the total historical cost of tangible fixed assets before accumulated depreciation, representing the firm's cumulative investment in productive physical assets. We construct the Capital Funding Prudence signal by calculating the year-over-year change in financing activities net cash flow, scaled by lagged gross property, plant, and equipment: $(\text{FINCF}(t) - \text{FINCF}(t-1)) / \text{PPEGT}(t-1)$. This ratio captures changes in a firm's external financing activities relative to its existing capital base, providing insight into how aggressively firms are altering their financing patterns. A higher value indicates an increase in net financing cash inflows relative to the firm's asset base, while a lower or negative value suggests reduced reliance on external financing or increased cash outflows to capital providers. We compute this measure using fiscal year-end values and require non-missing observations for both FINCF in consecutive years and lagged PPEGT to ensure meaningful comparisons. Observations with zero or negative PPEGT values are excluded from the analysis to avoid undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CFP signal. Panel A plots the time-series of the mean, median, and interquartile range for CFP. On average, the cross-sectional mean (median) CFP is 0.53 (0.00) over the 1989 to 2024 sample, where the starting date is determined by the availability of the input CFP data. The signal’s interquartile range spans -0.45 to 1.55. Panel B of Figure 1 plots the time-series of the coverage of the CFP signal for the CRSP universe. On average, the CFP signal is available for 6.20% of CRSP names, which on average make up 7.29% of total market capitalization.

4 Does CFP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CFP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CFP portfolio and sells the low CFP portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CFP strategy earns an average return of 0.29% per month with a t-statistic of 3.40. The annualized Sharpe ratio of the strategy is 0.57. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.56 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.09,

with a t-statistic of 4.57 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 537 stocks and an average market capitalization of at least \$2,756 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 2.03. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 15-35bps/month. The lowest return, (15 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.70. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CFP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twelve cases.

Table 3 provides direct tests for the role size plays in the CFP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CFP, as well as average returns and alphas for long/short trading CFP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CFP strategy achieves an average return of 20 bps/month with a t-statistic of 1.90. Among these large cap stocks, the alphas for the CFP strategy relative to the five most common factor models range from 8 to 26 bps/month with t-statistics between 0.76 and 2.40.

5 How does CFP perform relative to the zoo?

Figure 2 puts the performance of CFP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CFP strategy falls in the distribution. The CFP strategy’s gross (net) Sharpe ratio of 0.57 (0.49) is greater than 94% (99%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CFP strategy (red line).² Ignoring trading costs, a \$1 invested in the CFP strategy would have yielded \$1.98 which ranks the CFP strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CFP strategy would have yielded \$1.51 which ranks the CFP strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CFP relative to those. Panel A shows that the CFP strategy gross alphas fall between the 66 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198906 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CFP strategy has a positive net generalized alpha for five out of the five factor models. In these cases CFP ranks between the 80 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does CFP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CFP with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CFP or at least to weaken the power CFP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CFP}CFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CFP. Stocks are finally grouped into five CFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CFP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CFP signal in these Fama-MacBeth regressions exceed 0.50, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on CFP is -0.12.

Similarly, Table 5 reports results from spanning tests that regress returns to the CFP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CFP strategy earns alphas that range from 17-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.05, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CFP trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.17.

7 Does CFP add relative to the whole zoo?

Finally, we can ask how much adding CFP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the CFP signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CFP is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$31.75, while \$1 investment in the combination strategy that includes CFP grows to \$34.99.

8 Conclusion

This study provides robust evidence that Capital Funding Prudence (CFP) represents a meaningful signal for predicting cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CFP generates economically and statistically significant abnormal returns that persist even after controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CFP delivers substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.57 and a net Sharpe ratio of 0.49 after accounting for transaction costs. The signal’s predictive power remains robust when evaluated against the Fama-French five-factor model augmented with momentum, generating monthly abnormal returns of 22 basis points gross (19 basis points net) with t-statistics exceeding conventional significance thresholds. Particularly noteworthy is the signal’s ability to generate alpha even after controlling for six closely related strategies from the factor zoo, including net external financing, asset growth, and changes in balance sheet composition, with a monthly alpha of 18 basis points (t-statistic = 2.17).

These results have important practical implications for both academic researchers

and investment practitioners. The persistence of CFP’s predictive power after controlling for transaction costs suggests that the signal may be implementable in real-world portfolio strategies. Furthermore, the fact that CFP generates alpha beyond existing financing and growth-related anomalies indicates that it captures a distinct dimension of firm behavior related to capital allocation decisions that is not fully explained by previously documented factors.

However, several limitations of this study warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is confined to U.S. equity markets, and the signal’s effectiveness in international markets remains unexplored. Second, the economic mechanisms underlying CFP’s predictive power require further theoretical development to fully understand why markets appear to misprice firms based on their capital funding prudence.

Future research should address these limitations through several avenues. International replication studies would establish whether CFP represents a global phenomenon or is specific to U.S. market structures. Time-varying analysis of the signal’s effectiveness across different market regimes and economic cycles would provide insights into when CFP is most valuable for prediction. Additionally, investigating the behavioral or institutional frictions that allow this anomaly to persist would contribute to our understanding of market efficiency and price formation. Finally, examining the interaction between CFP and corporate governance characteristics might reveal whether the signal’s effectiveness varies with firm-level monitoring and control mechanisms.

In conclusion, Capital Funding Prudence emerges as a robust and economically meaningful predictor of cross-sectional stock returns that adds value beyond existing factor models and related anomalies, warranting its consideration in both academic asset pricing models and practical investment strategies.

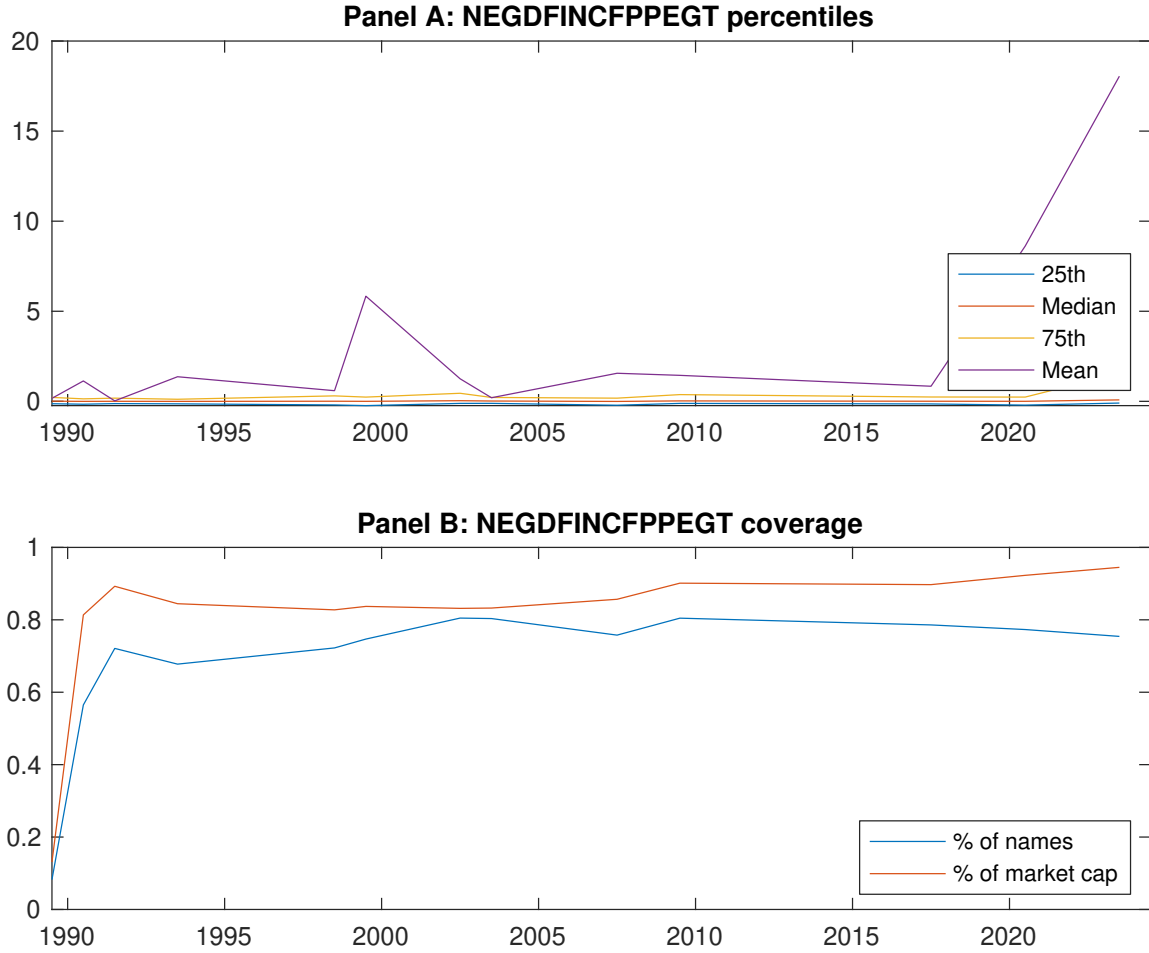


Figure 1: Times series of CFP percentiles and coverage.
This figure plots descriptive statistics for CFP. Panel A shows cross-sectional percentiles of CFP over the sample. Panel B plots the monthly coverage of CFP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CFP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Excess returns and alphas on CFP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.61 [2.37]	0.63 [2.96]	0.73 [3.93]	0.82 [3.94]	0.90 [3.71]	0.29 [3.40]
α_{CAPM}	-0.25 [-3.49]	-0.07 [-1.19]	0.15 [1.92]	0.14 [2.14]	0.10 [1.44]	0.34 [4.03]
α_{FF3}	-0.21 [-3.30]	-0.08 [-1.26]	0.11 [1.60]	0.13 [2.09]	0.13 [2.24]	0.34 [3.99]
α_{FF4}	-0.15 [-2.38]	-0.05 [-0.88]	0.11 [1.61]	0.10 [1.55]	0.12 [2.02]	0.27 [3.18]
α_{FF5}	-0.09 [-1.48]	-0.15 [-2.47]	-0.02 [-0.35]	-0.00 [-0.02]	0.18 [2.96]	0.27 [3.14]
α_{FF6}	-0.05 [-0.81]	-0.13 [-2.10]	-0.01 [-0.19]	-0.02 [-0.35]	0.17 [2.76]	0.22 [2.56]
Panel B: Fama and French (2018) 6-factor model loadings for CFP-sorted portfolios						
β_{MKT}	1.07 [72.41]	0.97 [63.10]	0.88 [53.20]	1.00 [66.58]	1.06 [69.14]	-0.01 [-0.63]
β_{SMB}	0.05 [2.55]	0.06 [2.83]	-0.12 [-4.98]	-0.03 [-1.51]	0.07 [3.31]	0.02 [0.61]
β_{HML}	-0.08 [-3.10]	-0.05 [-1.95]	0.02 [0.85]	-0.11 [-4.36]	-0.14 [-5.11]	-0.06 [-1.53]
β_{RMW}	-0.12 [-4.49]	0.14 [5.05]	0.19 [6.40]	0.19 [6.81]	-0.10 [-3.71]	0.02 [0.46]
β_{CMA}	-0.25 [-6.53]	0.08 [1.96]	0.21 [5.08]	0.20 [5.30]	-0.02 [-0.45]	0.23 [4.24]
β_{UMD}	-0.07 [-5.16]	-0.04 [-2.61]	-0.02 [-1.14]	0.03 [2.35]	0.02 [1.33]	0.09 [4.57]
Panel C: Average number of firms (n) and market capitalization (me)						
n	890	549	539	537	910	
me (\$10 ⁶)	2837	2756	3249	3543	2957	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CFP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [3.40]	0.34 [4.03]	0.34 [3.99]	0.27 [3.18]	0.27 [3.14]	0.22 [2.56]
Quintile	NYSE	EW	0.40 [5.30]	0.41 [5.48]	0.42 [5.62]	0.39 [5.19]	0.37 [5.15]	0.36 [4.91]
Quintile	Name	VW	0.26 [2.41]	0.29 [2.66]	0.29 [2.62]	0.22 [1.96]	0.23 [2.08]	0.18 [1.63]
Quintile	Cap	VW	0.39 [4.24]	0.44 [4.71]	0.43 [4.67]	0.36 [3.90]	0.34 [3.57]	0.28 [3.02]
Decile	NYSE	VW	0.24 [2.03]	0.29 [2.37]	0.28 [2.33]	0.19 [1.62]	0.25 [2.02]	0.18 [1.51]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [2.94]	0.29 [3.36]	0.28 [3.33]	0.24 [2.87]	0.23 [2.67]	0.19 [2.32]
Quintile	NYSE	EW	0.15 [1.70]	0.15 [1.62]	0.15 [1.63]	0.14 [1.51]	0.09 [1.02]	0.09 [1.04]
Quintile	Name	VW	0.22 [1.99]	0.23 [2.06]	0.22 [2.02]	0.18 [1.64]	0.18 [1.64]	0.15 [1.35]
Quintile	Cap	VW	0.35 [3.84]	0.39 [4.16]	0.38 [4.14]	0.34 [3.70]	0.30 [3.26]	0.27 [2.93]
Decile	NYSE	VW	0.20 [1.64]	0.22 [1.84]	0.22 [1.80]	0.16 [1.38]	0.19 [1.61]	0.15 [1.29]

Table 3: Conditional sort on size and CFP

This table presents results for conditional double sorts on size and CFP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CFP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CFP and short stocks with low CFP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198906 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CFP Quintiles					CFP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.41 [1.11]	0.93 [2.96]	1.08 [3.21]	0.92 [2.80]	0.87 [2.28]	0.46 [3.46]	0.46 [3.38]	0.47 [3.47]	0.43 [3.13]	0.41 [2.99]	0.39 [2.81]
	(2)	0.50 [1.53]	0.83 [2.77]	0.92 [3.33]	0.84 [2.89]	0.87 [2.65]	0.36 [3.00]	0.40 [3.25]	0.39 [3.22]	0.31 [2.55]	0.35 [2.91]	0.30 [2.44]
	(3)	0.65 [2.09]	0.80 [2.98]	0.88 [3.68]	0.81 [3.05]	0.95 [3.10]	0.30 [2.34]	0.32 [2.53]	0.33 [2.53]	0.27 [2.10]	0.38 [2.90]	0.34 [2.57]
	(4)	0.69 [2.30]	0.72 [2.90]	0.85 [3.90]	0.82 [3.37]	1.02 [3.76]	0.33 [2.81]	0.41 [3.49]	0.38 [3.30]	0.33 [2.88]	0.23 [1.97]	0.20 [1.74]
	(5)	0.67 [2.60]	0.51 [2.49]	0.75 [3.80]	0.82 [4.09]	0.87 [3.62]	0.20 [1.90]	0.26 [2.40]	0.25 [2.35]	0.16 [1.51]	0.15 [1.39]	0.08 [0.76]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CFP Quintiles					CFP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	365	366	367	365	361	42	44	42	43	41	
	(2)	112	113	113	113	112	78	80	80	80	78	
	(3)	78	79	79	79	78	141	141	141	141	139	
	(4)	68	68	68	68	67	314	315	314	319	312	
(5)	61	62	62	62	61	2048	2423	2864	2912	2209		

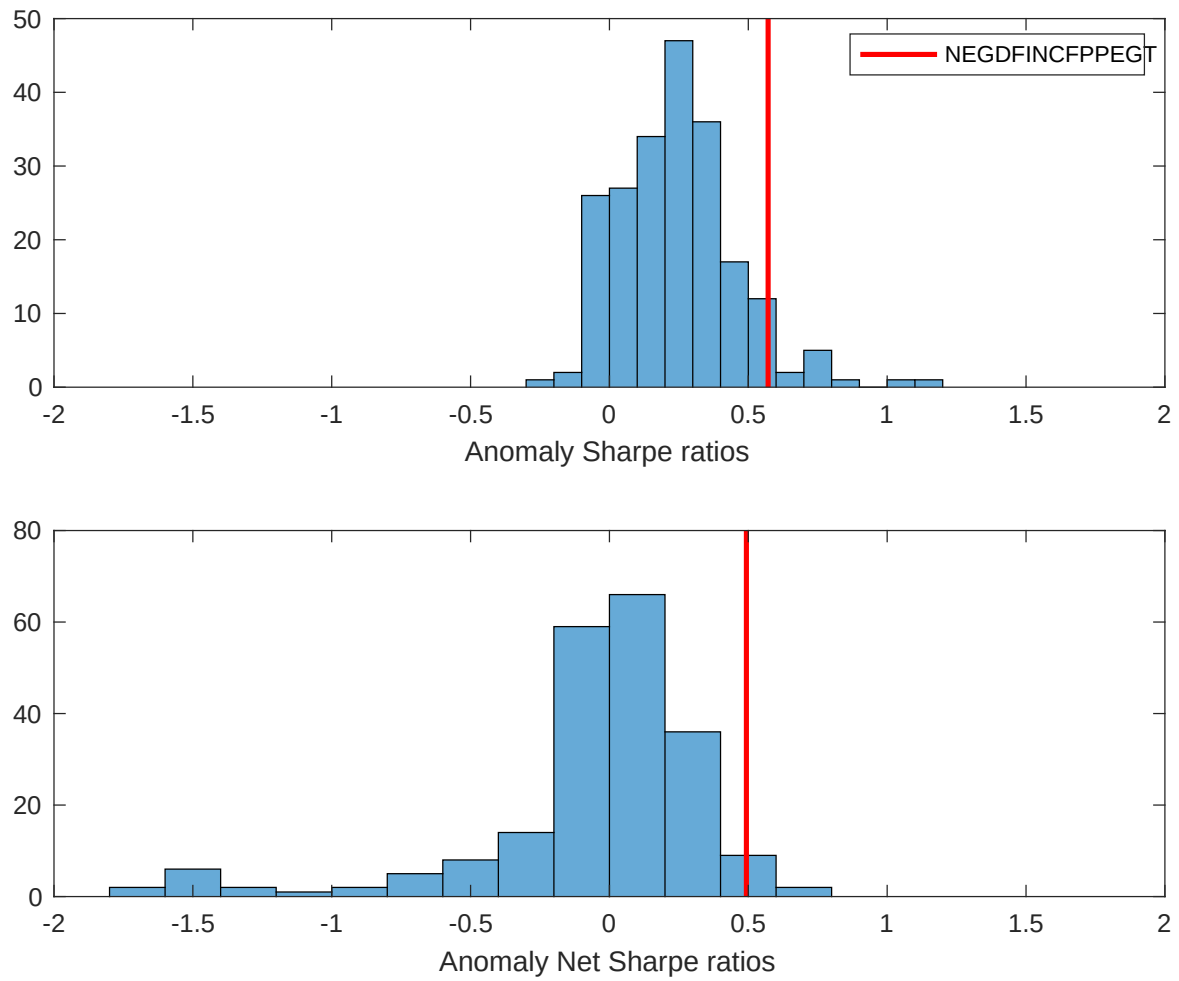


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CFP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

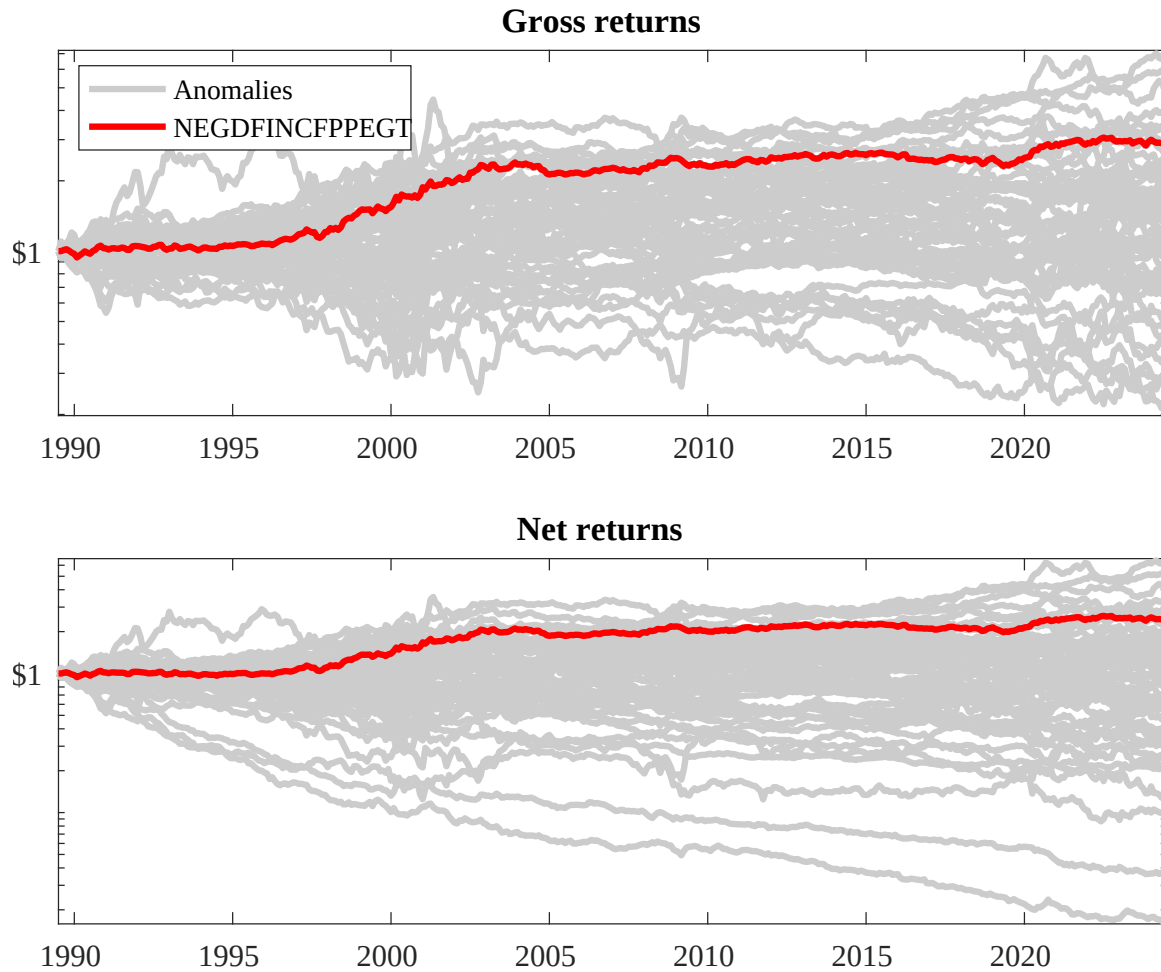
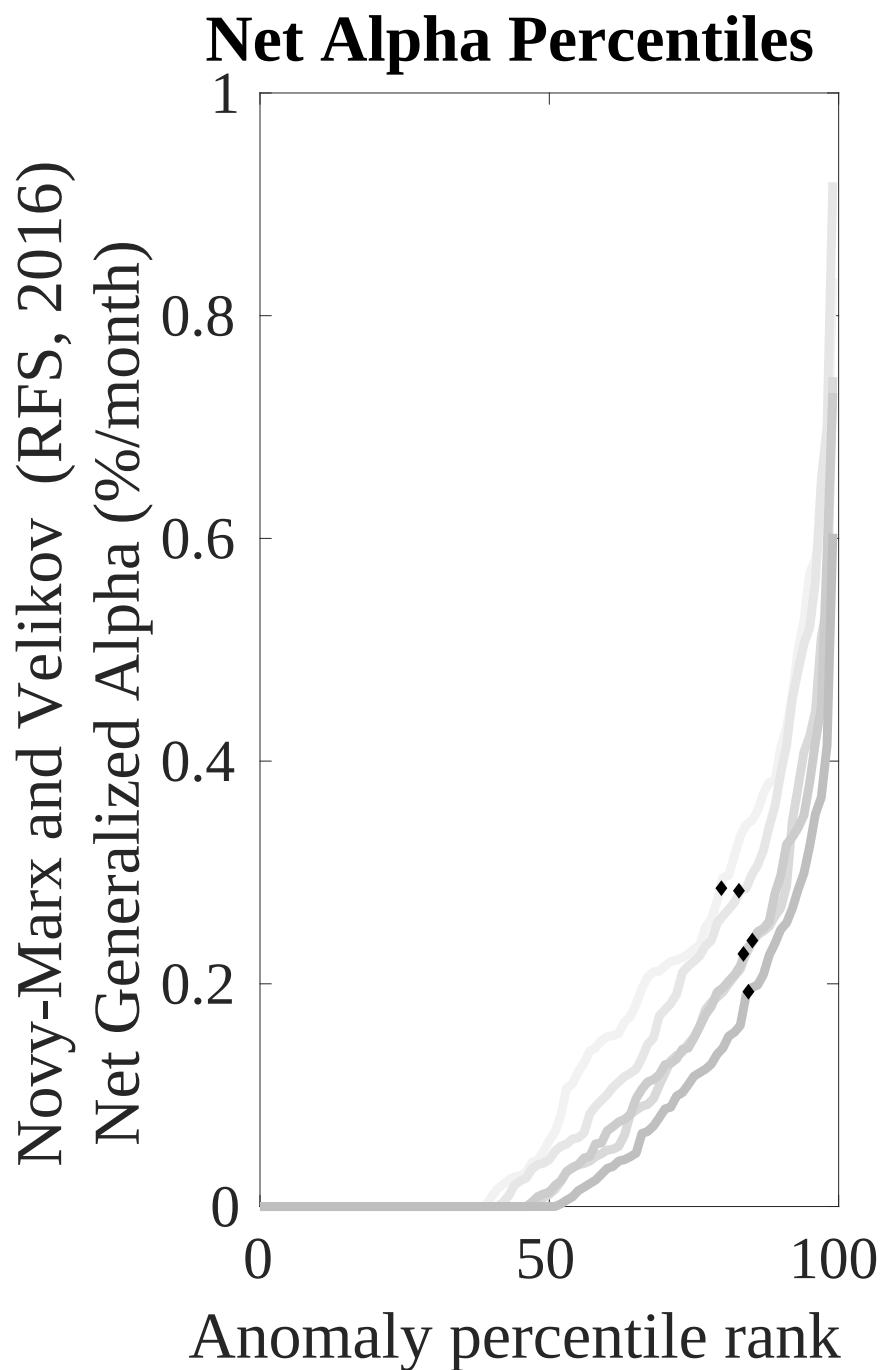
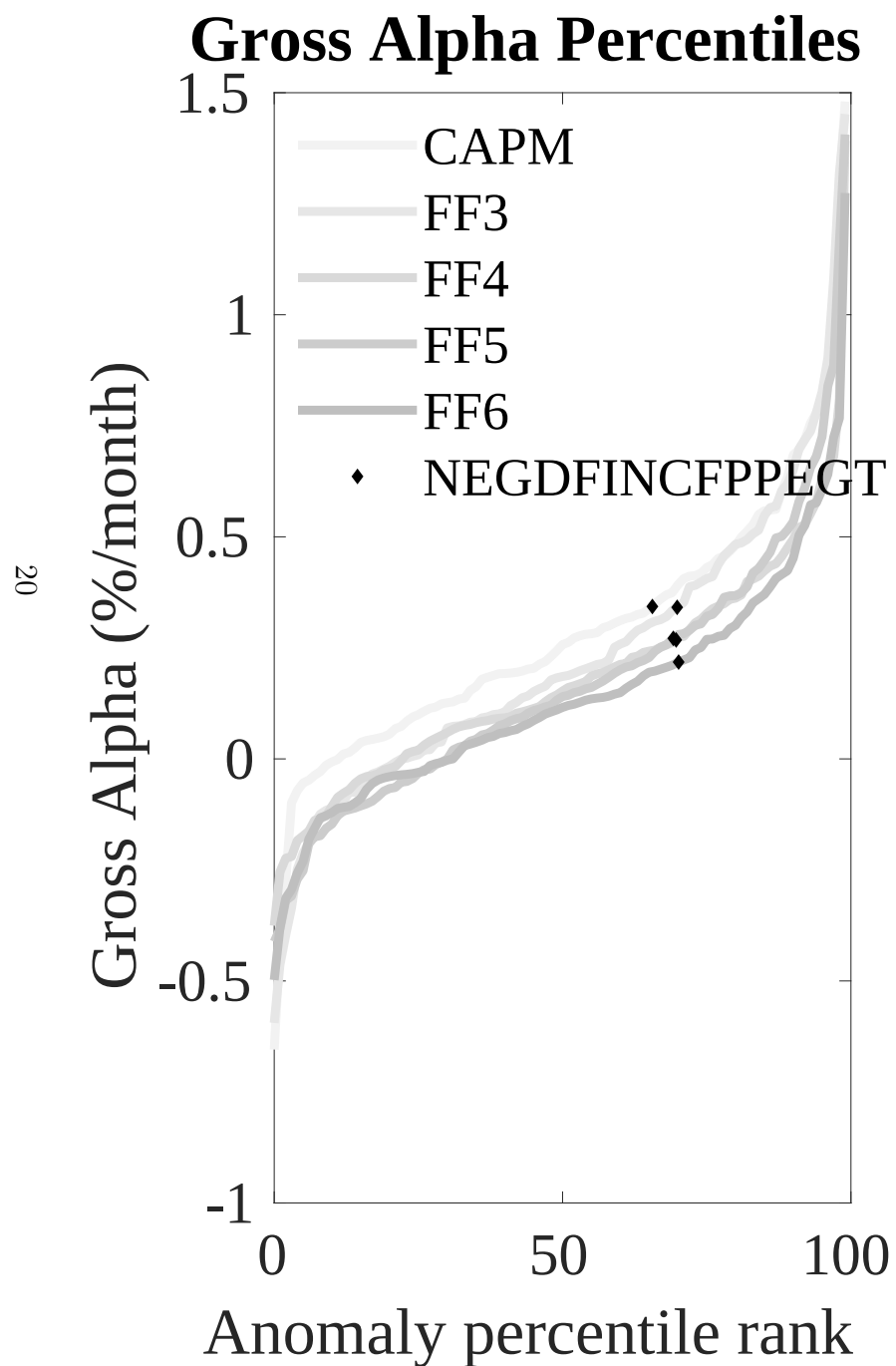


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CFP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



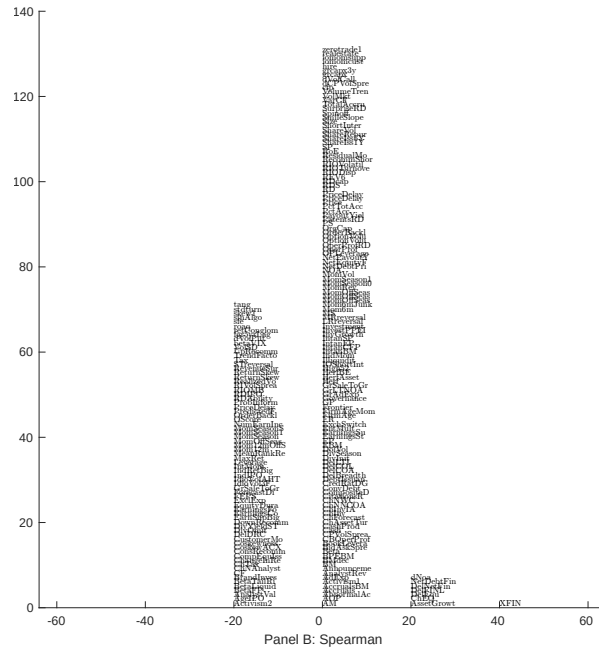
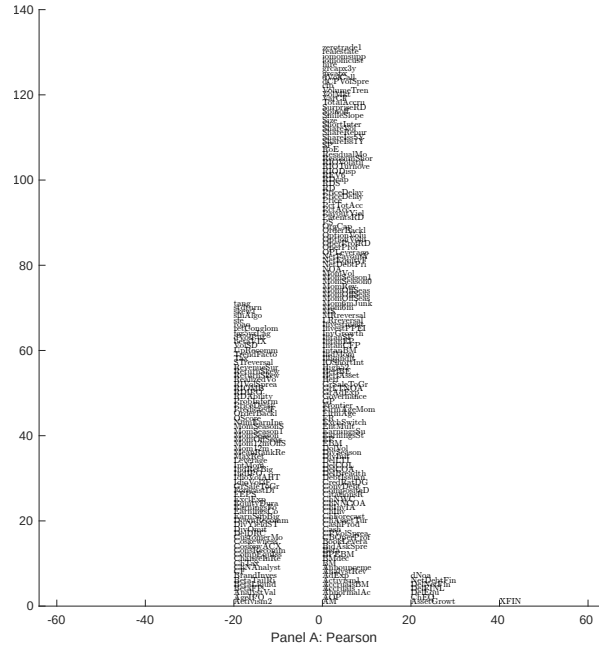


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with CFP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

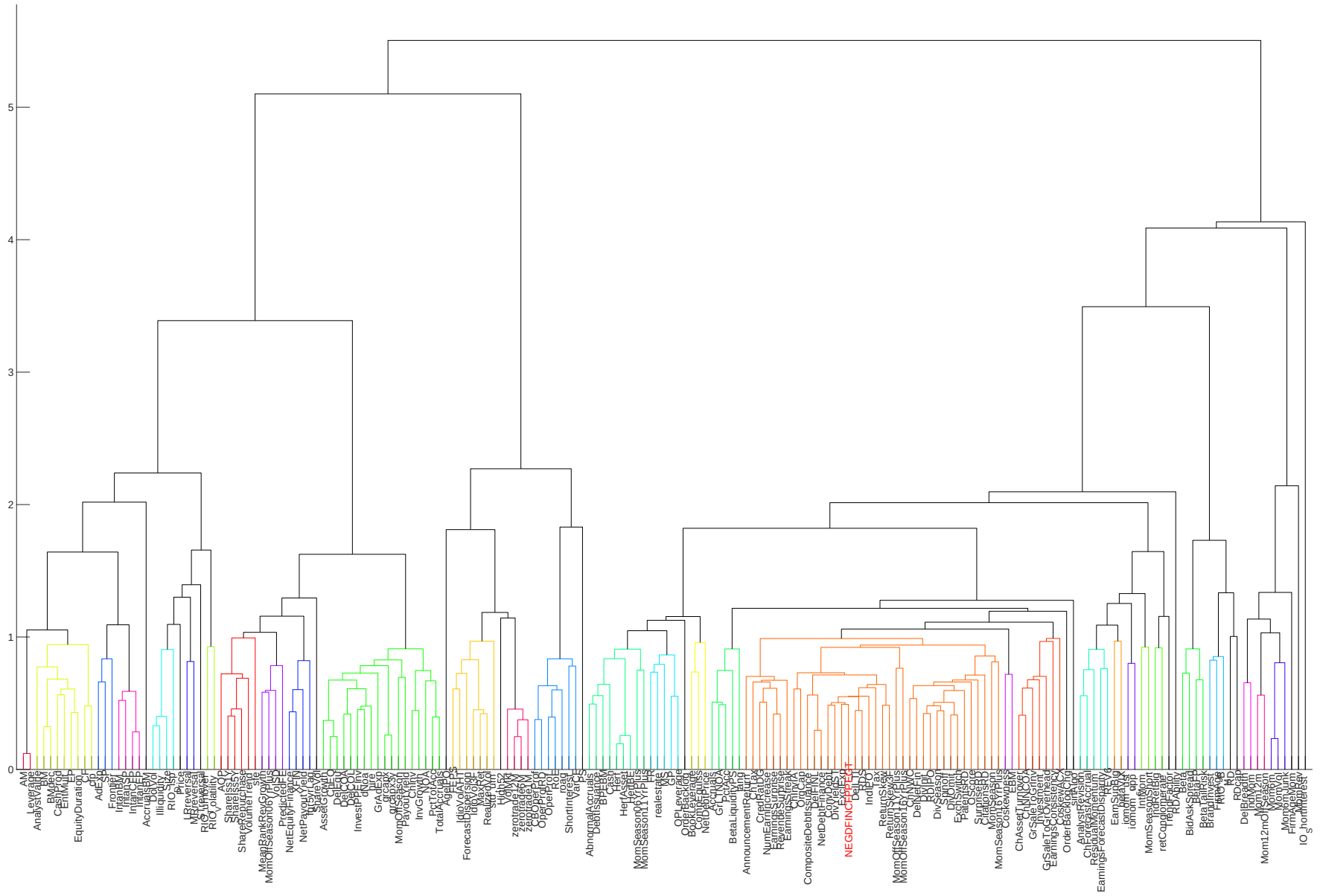


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

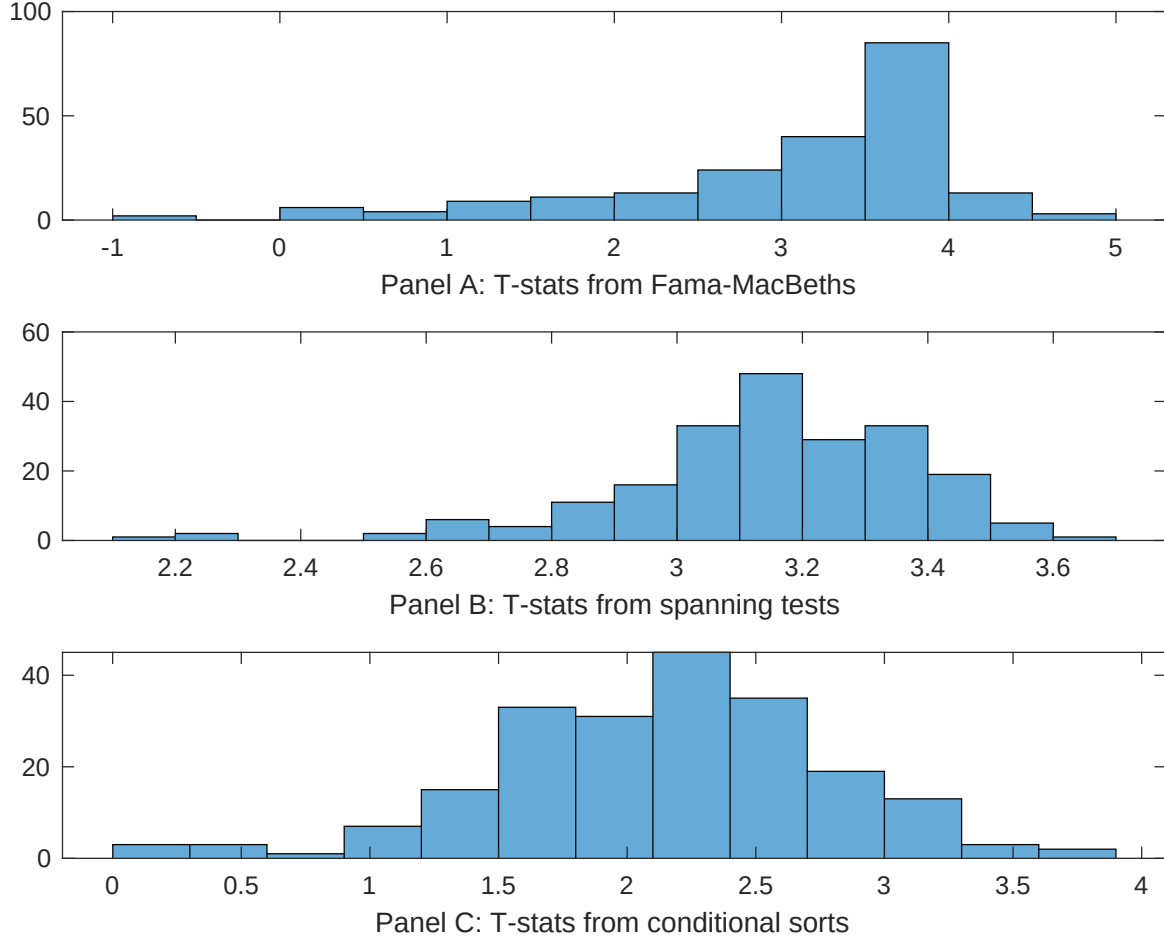


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CFP}CFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CFP. Stocks are finally grouped into five CFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CFP, and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CFP}CFP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.12 [4.35]	0.13 [4.28]	0.17 [5.58]	0.12 [3.89]	0.12 [4.13]	0.12 [3.94]	0.13 [4.93]
CFP	0.88 [0.62]	0.57 [0.50]	0.22 [1.88]	0.57 [4.05]	0.29 [2.52]	0.28 [2.59]	-0.20 [-0.12]
Anomaly 1	0.20 [4.83]						0.19 [3.85]
Anomaly 2		0.86 [7.37]					0.33 [0.17]
Anomaly 3			0.46 [6.08]				0.89 [0.80]
Anomaly 4				0.33 [5.40]			0.70 [1.00]
Anomaly 5					0.11 [7.64]		0.55 [2.38]
Anomaly 6						0.12 [4.13]	0.17 [0.34]
# months	420	420	420	420	420	420	420
$\bar{R}^2(\%)$	1	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CFP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CFP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.17 [2.05]	0.20 [2.33]	0.19 [2.28]	0.21 [2.50]	0.19 [2.23]	0.19 [2.29]	0.18 [2.17]
Anomaly 1	14.18 [3.46]						11.04 [2.58]
Anomaly 2		11.53 [2.59]					4.92 [0.81]
Anomaly 3			8.89 [1.95]				5.20 [0.72]
Anomaly 4				9.96 [3.26]			7.93 [2.51]
Anomaly 5					11.50 [2.46]		4.63 [0.92]
Anomaly 6						6.89 [1.58]	-4.91 [-0.66]
mkt	0.82 [0.38]	-1.14 [-0.54]	-0.92 [-0.43]	-1.09 [-0.52]	-1.44 [-0.69]	-1.15 [-0.54]	0.62 [0.29]
smb	6.82 [2.07]	1.54 [0.51]	1.57 [0.52]	2.94 [0.98]	2.20 [0.73]	2.15 [0.71]	5.96 [1.76]
hml	-4.23 [-1.15]	-6.93 [-1.88]	-6.69 [-1.81]	-6.60 [-1.81]	-7.13 [-1.93]	-6.71 [-1.81]	-5.79 [-1.56]
rmw	-5.93 [-1.32]	2.72 [0.72]	2.13 [0.56]	4.12 [1.08]	2.72 [0.71]	2.85 [0.75]	-3.13 [-0.68]
cma	14.02 [2.37]	9.51 [1.27]	14.90 [2.18]	14.62 [2.47]	14.42 [2.25]	16.27 [2.35]	-0.05 [-0.01]
umd	8.00 [4.33]	8.69 [4.67]	8.18 [4.38]	7.82 [4.21]	8.20 [4.41]	8.37 [4.49]	7.55 [4.05]
# months	420	420	420	420	420	420	420
$\bar{R}^2(\%)$	14	13	12	14	13	12	15

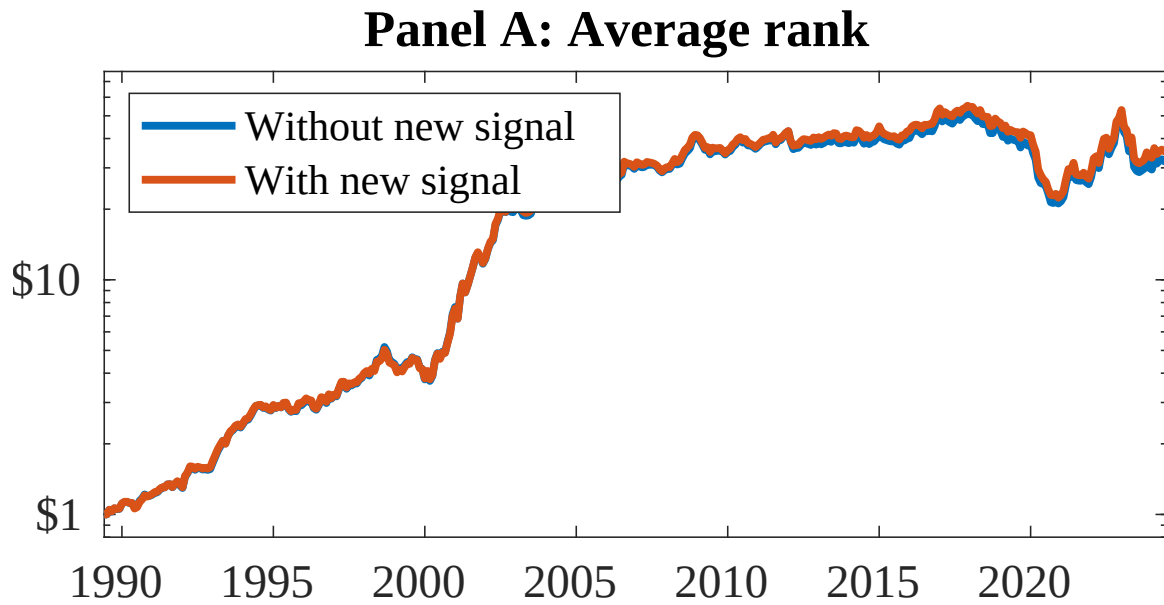


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as CFP. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Ai, H., Kiku, D., and Li, R. (2013). A mechanism design model of firm dynamics: The case of limited commitment. *Journal of Finance*, 68(6):2351–2393.
- Berk, J. B., Green, R. C., and Naik, V. (1999). Optimal investment, growth options, and security returns. *Journal of Finance*, 54(5):1553–1607.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Cooper, R. W. and Haltiwanger, J. C. (2006). On the nature of capital adjustment costs. *Review of Economic Studies*, 73(3):611–633.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Livdan, D., Sapriza, H., and Zhang, L. (2009). Financially constrained stock returns. *Journal of Finance*, 64(4):1827–1862.
- Lyandres, E., Sun, L., and Zhang, L. (2008). The new issues puzzle: Testing the investment-based explanation. *Review of Financial Studies*, 21(6):2825–2855.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Titman, S., Wei, K. C. J., and Xie, F. (2004). Capital investments and stock returns. *Journal of Financial and Quantitative Analysis*, 39(4):677–700.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.